GP FUTUREMAKERS 2026 · AI FOR SOCIAL GOOD

AI Shopkeeper Assistant — Social Impact & Business Report

Prepared for: GP FutureMakers 2026 — *"AI for Social Good"*
Date: 13 October 2026
Product status: working Android app (Flutter) on a live Supabase backend, with Gemini AI for Bangla voice and camera selling. Tested internally; no external pilot yet.

> All money figures are in Bangladeshi Taka (৳) at **৳123 = US$1** (Bangladesh Bank reference rate, July 2026). AI costs are **estimates** built from the app's actual code and Google's published prices. Each calculation states its assumptions so a judge can check it.

Contents

Every entry is a link. Section references like §6.1 in the text are links too, and each section heading has a link back to this page.

1. Summary

2. The problem

- 2.1 Who we serve
- 2.2 What goes wrong today
- 2.3 Women are hit hardest

3. The solution: what the app does today

- 3.1 AI features: the core of the social case
- 3.2 Bookkeeping features
- 3.3 Not built yet (stated honestly)

4. Market analysis

- 4.1 Market size
- 4.2 Why now
- 4.3 Competitors

5. Social impact

- 5.1 Mapping to the FutureMakers challenge areas
- 5.2 Theory of change
- 5.3 How we will measure impact
- 5.4 Responsible AI

6. Business model & financials

- 6.1 What it costs to run
- 6.2 Pricing
- 6.3 Unit economics (one Smart shop, optimised AI)
- 6.4 Three-year scenario (illustrative, not a forecast)
- 6.5 Other revenue streams (Phase 2+)

7. Go-to-market plan

- 7.1 Partnership ideas for Grameenphone

8. Risks and how we handle them

9. What we need

10. Conclusion

Sources

1. Summary

Back to contents

Bangladesh's neighbourhood *mudi dokans* still run on a paper khata. Stock is counted by eye, nobody knows the day's real profit, and *baki* (customer credit) is a column of pencil marks that is hard to total and easy to dispute. Free digital khata apps exist. But they expect the shopkeeper to **type**, and **42% of Bangladeshis are below fifth-grade literacy**.

AI Shopkeeper Assistant lets a shopkeeper make a sale by speaking Bangla or by pointing the camera at the goods. Behind that, it keeps stock, profit and customer credit correct automatically. A spoken sentence like "*আড়াই কেজি চাল আর দুইটা পেপসি*" ("two and a half kilos of rice and two Pepsis") becomes a checked sale. Stock falls, profit is recorded, and the customer's baki updates if it was a credit sale.

Social problem	Low-literacy, low-margin shopkeepers (many of them women) can't use typing-heavy tools. Without records they can't see profit, manage stock, collect baki or show a lender any history
Our answer	Voice-first, camera-first AI bookkeeping in Bangla. The AI only *suggests*; the shopkeeper confirms every sale
Market	1–4.5 million small shops (estimates vary); 73.4% of households own a smartphone; Bangla QR became mandatory for merchants in July 2026, so shops are being pushed to go digital right now
Business model	Free core app, plus a paid **Smart plan (৳299/month)** for AI voice and camera, then sponsored and lending partnerships
Unit economics	About ৳0.20 of AI cost per voice or camera sale after one planned optimisation, which gives a ~55% contribution margin. **Break-even at ~150 paying shops**
FutureMakers fit	Digital access & education (primary); gender equality & inclusion; skills & jobs; safety, security & data privacy

2. The problem

Back to contents

2.1 Who we serve

The **mudi dokan**, a 100–200 sq ft neighbourhood grocery holding 500–1,000 different items (Khulna grocery study). These shops carry almost all of Bangladesh's everyday consumption.

- ShopUp estimates **4.5 million small shops account for 98% of consumption** (Dhaka Tribune). This is a company estimate.
- The **Economic Census 2024** counted **11.70 million economic units**, with wholesale & retail trade (plus vehicle repair) the largest group at **41.82%** (UNB).
- Older sector estimates put grocery shops alone at **about 1 million** (industry overview).

There is no official count of grocery shops alone, so this report uses **1–4.5 million** as the range.

2.2 What goes wrong today

Pain	Consequence
No stock record	Shopkeepers find out an item is gone when a customer asks for it, and over-order slow sellers. Working capital is tied up in the wrong goods
No profit visibility	Margins are thin (around 3–4% net at small retail shops in one study). A shop can be busy and still lose money without noticing
Paper baki	Credit given to neighbours is hard to total and easy to dispute. Money owed quietly goes uncollected
No financial history	Banks lend on records and collateral. **More than 60% of micro-enterprises cannot access credit**, and those that can often pay **24–40% interest** (TBS)
Typing is a barrier	**42.45% of people are below fifth-grade literacy** (Bonik Barta, HMSS 2025); functional literacy is **62.92%** (Prothom Alo, BBS). Apps that need typing leave out the shopkeepers who need help most

2.3 Women are hit hardest

Women-led enterprises hold only **7.28% of small-business loans**, against Bangladesh Bank's **15% target**, as of March 2026 (Dhaka Tribune). The reasons are collateral rules (few women hold property titles) and paperwork such as audited statements. A clean digital sales history is the kind of record that can replace some of that paperwork.

3. The solution: what the app does today

Back to contents

Everything in this section is **built and tested** (443 automated tests passing).

3.1 AI features: the core of the social case

Feature	What the shopkeeper does	Why it matters socially
Voice sale (Bangla, English, Banglish)	Says the sale out loud, e.g. *"arai kg khola chal, dui ta pepsi"*. The AI turns it into items, quantities and products	No typing. Understands Bengali fraction words such as আধা (½), দেড় (1½), আড়াই (2½), সাড়ে, সোয়া and পৌনে, and checks them in code so a spoken "2½ kg" is never recorded as "2 kg"
Camera scan session	Points the camera at the items and taps once per item. Recognition runs in the background, and everything is reviewed in one tray	A ten-item basket goes from about **40 taps / 75 seconds** in the old one-photo-at-a-time flow to about **11 taps / 20 seconds** (our design estimate, to be confirmed in the pilot). That's faster checkout with no typing
AI never decides alone	Every AI result appears on a confirmation screen before anything is saved	Protects shopkeepers from AI mistakes. Required by our own design rules
Smart tie-breaking	If "Pepsi" could mean 500 ml or 1 L, the app asks, or leans towards the one this shop actually sells most	Fewer wrong sales, and less frustration for low-literacy users

3.2 Bookkeeping features

- **Inventory:** products with categories, units (pieces, kg, g, litre, ml), buying and selling prices, archived products, and the total value of stock.
- **Cart and checkout:** several items per sale, cash or **baki (credit)**, and a flat discount.
- **Stock can never go wrong:** each sale updates stock and records the sale in one step on the server, so stock can't go negative and a sale can't be half-saved.
- **True profit:** each sale freezes the buying and selling price at that moment, so past profit stays correct when prices change.
- **Customer khata:** a running credit ledger per customer, recording payments, showing who owes what, and a total for money owed to the shop.
- **Dashboard and reports:** today's and weekly revenue and profit, trend charts, a breakdown for any day, and payment-method splits.
- **Sales history:** a calendar filter and full details of every past sale.
- **Full Bengali interface:** an in-app language switch, Bengali numerals and a Bengali font.
- **Privacy by design:** each shop can only ever see its own data, enforced by the database itself (Row Level Security).

3.3 Not built yet (stated honestly)

Gap	Status
AI insights & trends screens	The screens exist, but the AI that writes the insights was never built. Roadmap
Working offline	Every sale currently needs internet. The app shows a "no internet" banner but can't record a sale offline. **This is the most important gap** (see §8)
Bangla QR / bKash payment recording	Not built. Roadmap
Low-stock push notifications	Not built

4. Market analysis

Back to contents

4.1 Market size

Layer	Estimate	Basis
Total market	**1–4.5 million** small shops	Range of the published estimates in §2.1
Reachable now	**~2–3 million**	Shop owners with a smartphone and mobile internet: **65.6%** of individuals own a mobile phone and **73.4%** of households own a smartphone (UNB/BBS, Dhaka Tribune/BBS); **135.9 million** internet subscriptions in June 2026 (Financial Express/BTRC)
3-year target	**10,000 paying shops**	About 0.3–0.5% of reachable shops. Deliberately modest for a student-founded venture

4.2 Why now

1. **Bangla QR became mandatory on 1 July 2026.** Every merchant must accept QR payments, and **2.4 million merchants** were registered by August (Observer/Bangladesh Bank). Shopkeepers are being pushed into digital tools now.
2. **Smartphones are already in most shops.** Household smartphone ownership is 73.4%.
3. **AI is now cheap enough.** A voice sale costs about ৳**0.2–0.3** in AI fees (§6). Two years ago this product would not have been affordable.
4. **A new data law makes trust a selling point.** The **Personal Data Protection Ordinance 2025** sets consent and data-handling rules (Daily Star). Shops will prefer tools that are built to comply.

4.3 Competitors

Competitor	What they offer	Strength	Where we differ
TallyKhata (Progoti Systems)	Free digital khata, sales/stock records, TallyPay wallet, QR payments	**1 crore+ downloads**, about 1M monthly users, **works offline**, central-bank-licensed wallet (TallyKhata)	Built around typing. No AI voice or camera selling
Hishabee	POS, inventory, accounting, **inventory finance up to ৳50,000** through Prime Bank	300,000+ businesses (homepage); claims 1M+ merchants reached (Hishabee, Impact Report)	Typing-based POS. We add voice and camera, and could integrate rather than compete
ShopUp / Mokam	B2B supply to shops plus "Baki" stock credit	Large distribution network, $30M debt financing (TBS)	A supply business, not shop bookkeeping. A possible partner
Hishab / Verbex.ai	Bangla voice AI over phone calls	Backed by Startup Bangladesh (৳2 crore) (Startup Bangladesh)	Enterprise call automation, not a shop app. Shows that Bangla voice AI is investable
Paper khata	Notebook	Free, universal, trusted	No totals, no profit, no history

Our position: *the shop app you can talk to.* The incumbents proved shopkeepers will adopt a digital khata. We reach the shopkeepers they leave out, the ones who won't or can't type, and we make every user faster at the counter.

5. Social impact

Back to contents

5.1 Mapping to the FutureMakers challenge areas

Challenge area	How the app contributes	Evidence it matters
Digital access & education *(primary)*	Speaking Bangla replaces typing, and the camera replaces searching lists. A shopkeeper with little schooling can still keep accurate digital books	42.45% below fifth-grade literacy; functional literacy 62.92%
Gender equality & inclusion	Women running home-based or small shops get a record of their business without relying on a literate relative to keep books. That record can support a loan application	Women-led firms hold 7.28% of small-business loans against a 15% target

Challenge area	How the app contributes	Evidence it matters
Skills development & jobs	Shopkeepers and their helpers learn digital bookkeeping, and a helper can record sales by voice. Profit and stock data help shops survive and keep employing people	60%+ of micro-enterprises can't access credit
Safety, security & data privacy	Each shop's data is locked to that shop by the database; the AI can't save anything without the shopkeeper's confirmation; we have a compliance plan for the 2025 data law (§8)	PDPO 2025 obligations on consent and data handling

Sustainable Development Goals: SDG 1 (No Poverty), SDG 5 (Gender Equality), SDG 8 (Decent Work & Economic Growth), SDG 9 (Industry & Innovation), SDG 10 (Reduced Inequalities).

5.2 Theory of change

1. Input — Bangla voice + camera AI app

→

2. Activity — Shopkeeper records every sale by voice or camera

→

3. Output — Accurate stock, profit and baki ledger

→

4. Outcome — Fewer stock-outs, baki collected, better buying; a sales history a lender can read

→

5. Impact — Higher, steadier shop income; credit access; women-run shops become financially visible

5.3 How we will measure impact

Promising impact isn't enough, so the pilot (§7) will measure it. These indicators come straight from data the app already records, plus a short survey at sign-up and after 3 months.

Outcome	Indicator	Source
Inclusion	% of users who sell mainly by voice or camera; % women owners; % owners below fifth-grade schooling	Sign-up survey + app usage
Time saved	Seconds per checkout, before and after	Timed observation + app data
Baki recovered	৳ of customer credit collected vs. outstanding, month by month	Khata ledger
Profit awareness	% of shops that check profit weekly; owners who can state their monthly profit	App usage + survey
Stock control	Number of times a top-selling item runs out	Inventory data
Financial access	Shops that use their app record for a loan or supplier credit	Partner data + survey

5.4 Responsible AI

- **A person always confirms.** No AI output is saved without the shopkeeper approving it.
- **The AI can't invent products.** It can only pick from the shop's own list; anything else is thrown away by the server.
- **Mistakes fail safely.** If the AI is unsure, it asks rather than guesses. If the AI service is down, the shopkeeper sees an error and can sell manually, rather than seeing a wrong result.

- **Shop data stays private.** We use Google's paid AI service, where business data is not used to train Google's models (the free tier is for testing only).

6. Business model & financials

Back to contents

6.1 What it costs to run

There are three running costs: **AI (Gemini)**, **server (Supabase)** and **internet**.

AI cost per voice or camera sale (estimate, ৳)

The app currently sends the shop's **whole product list** to the AI with every request, so cost grows with shop size:

Shop size	Voice sale	Camera scan, today's model (2026 price)	Camera scan, today's model (**from Jan 2027**, price doubles)	Camera scan on the cheaper model
100 products	৳0.27	৳0.38	৳0.76	৳0.19
300 products	৳0.44	৳0.77	৳1.54	৳0.34
1,000 products	৳1.06	৳2.13	৳4.25	৳0.89

Assumptions: Google's published prices (Gemini API pricing): Flash-Lite (voice) $0.30 / $2.50 per million input / output tokens; 3.6 Flash (scan) $0.75 / $3.75, rising to $1.50 / $7.50 on 1 Jan 2027. About 21–24 tokens per product in the list, 32 tokens per second of audio (Gemini docs), an ~8-second voice clip, and 300–400 output tokens.

The key fix: send a shortlist, not the whole shop. If the app sends the AI only the ~60 most likely products (recent sellers plus a name match), and uses the cheaper Flash-Lite model for the camera too, **every AI sale costs about ৳0.20 whatever the shop's size.** This is a few days of engineering, and it's needed before charging (see §8).

AI sales per month	Optimised (~৳0.20 each)	Today, 300-item shop	Today, 300-item shop, 2027 prices
300 (10/day)	৳59	৳182	৳297
600 (20/day)	৳117	৳364	৳595
900 (30/day)	৳176	৳546	৳892

Server: Supabase Pro costs **$25 ≈ ৳3,075 a month** and includes 2 million AI-function calls (Supabase pricing guide). That's enough for thousands of shops, so it works out at **under ৳10 per shop**. The free plan is fine for the pilot.

Internet: the shopkeeper uses their own mobile data. One camera scan uploads about 50–70 KB and one voice sale about 100–300 KB, so 20 AI sales a day uses only **about 100–200 MB a month**. Manual sales use almost nothing.

6.2 Pricing

Plan	Price	Includes	Who it's for
Free	৳0	Inventory, khata, cart, history, dashboard, manual and quick-tap selling, plus **30 AI sales a month** to try it	Every shop. Builds habit and trust
Smart	৳**299 / month**, paid by bKash/Nagad	**600 AI voice and camera sales a month** (20 a day), priority support	Busy shops that want speed
AI top-up	৳40 per 100 extra AI sales	Prepaid, like a mobile recharge	Heavy users. Familiar pay-as-you-go model
Sponsored *(Phase 2)*	Paid by a partner	The Smart plan, free to the shop	Distributors, FMCG brands or lenders paying for their retailer network

Why ৳299: that's about ৳10 a day, less than one cup of tea at the counter. Saving one forgotten baki payment of a few hundred taka covers the whole month.

6.3 Unit economics (one Smart shop, optimised AI)

	৳ / month
Price	299
AI cost (600 sales × ৳0.195)	−117
Payment fee (assumed ~2%)	−6
Server share	−10
Contribution per shop	**≈ 166 (55%)**

Break-even: fixed costs of about ৳**25,000 a month** (server ৳3,075; one part-time field agent and marketing ৳20,000; tools ৳2,000) ÷ ৳166 = **about 150 paying shops.**

6.4 Three-year scenario (illustrative, not a forecast)

	Year 1	Year 2	Year 3
Paying shops, end of year	200	2,000	10,000
Revenue	৳3.6 lakh	৳39.5 lakh	৳2.15 crore
Contribution (55%)	৳2.0 lakh	৳21.9 lakh	৳1.19 crore

Assumes ৳299/month and an average of the shop count at the start and end of each year. This leaves out sponsored plans and lending referrals.

6.5 Other revenue streams (Phase 2+)

1. **Lending referrals:** with the shopkeeper's consent, share their sales history with a bank or microfinance partner and earn a referral fee per loan. Hishabee's finance partnership with Prime Bank shows banks want this kind of data.

2. **Sponsored retailer networks:** an FMCG company or distributor pays for the Smart plan for its retailers. In return it gets anonymised, consented sell-through data and faster restock orders.
3. **Supplier ordering:** a low-stock item turns into a restock order to a wholesaler, and we take a small commission.

7. Go-to-market plan

Back to contents

Phase	Timeline	Goal	Key actions
0. Fix & pilot	Months 0–3	30–50 shops, impact data	Build offline manual sales and the AI shortlist (§8); give the app free to shops near campus; measure the §5.3 indicators
1. Launch	Months 3–12	1,000 users, 200 paying	Launch the Smart plan; field demos at the counter (voice is best shown live); shop-to-shop referrals (one free month per referral)
2. Partner	Year 2	2,000 paying	Sponsored plans with a distributor or FMCG brand; lending partner; Bangla QR payment recording
3. Scale	Year 3	10,000 paying	AI insights; supplier ordering; expansion beyond grocery shops (pharmacies, hardware)

7.1 Partnership ideas for Grameenphone

1. **"Dokan data pack":** GP bundles or zero-rates the app's data for shopkeepers, which removes the internet-cost worry.
2. **AI top-ups through GP recharge:** shopkeepers already understand prepaid recharge, so AI credit can work the same way.
3. **GP's retail network as the pilot channel:** GP reported **more than 350,000 points of sale** in 2017 (Telenor CMD 2017; current figure not found). Many of them are mudi dokans that already know GP.
4. **Alignment with GP's "AI & I" programme** to become an AI-native telco (Daily Star).

8. Risks and how we handle them

Back to contents

Risk	Impact	Mitigation
Needs internet for every sale	High. TallyKhata works offline, and a failed sale with a customer waiting loses trust	**Before the pilot:** let manual sales be saved offline and synced later, with AI used only when online
AI cost grows with shop size	High. A 1,000-item shop can cost ৳1–4 per AI sale	**Before charging:** send a ~60-product shortlist and use the cheaper model for camera scans (§6.1). Caps and prepaid top-ups protect the margin
AI price rise on 1 Jan 2027	Medium. Today's camera model doubles in price	Move camera scans to Flash-Lite, which has no scheduled increase
Strong free competitors	Medium	Compete on what they lack (voice and camera), and stay open to integrating with them

Risk	Impact	Mitigation
AI misreads a sale	Medium	Confirmation is required before saving; the AI can only pick from the shop's own products; stock is updated safely on the server
Data protection law	Medium	Clear Bangla consent at sign-up, a privacy policy, data-retention rules. **Our database currently runs in Tokyo.** We will review the law's cross-border and local-storage rules with a lawyer and move to the nearest compliant region if needed
Adoption and trust	Medium	Live demos at the counter, a free tier, referrals, and a Bangla-first design
Team and ownership	Medium	A written agreement among the three founders on ownership and roles before any revenue

9. What we need

Back to contents

Need	Use
Mentorship (FutureMakers development phase)	Pricing validation, pilot design, telco partnership
~৳1.5 lakh pilot budget	Server and AI for 50 shops for 3 months (under ৳20,000), field visits and printed Bangla guides, survey work
Grameenphone partnership	A data bundle and access to GP retailers for the pilot

10. Conclusion

Back to contents

Bangladesh has millions of small shops and a ready-made push to go digital through the Bangla QR mandate. Existing tools mostly serve shopkeepers who are comfortable typing. **AI Shopkeeper Assistant brings proper bookkeeping (stock, profit, baki and a history a lender can read) to shopkeepers who would rather talk.** The core app is built and tested. AI costs about ৳0.20 per sale once the planned optimisation is done, and the business breaks even at about 150 paying shops. The next step is a 50-shop pilot that measures the impact rather than assuming it.

Sources

Back to contents

Competition

- Grameenphone launches second FutureMakers competition — The Daily Star
- Grameenphone invites students to FutureMakers AI competition — TBS
- FutureMakers returns for its 2nd edition — Prothom Alo

Market & connectivity

- Economic Census 2024 report — UNB
- ShopUp secures $30m debt financing — Dhaka Tribune
- Overview of Bangladesh Retail Industry — Scribd
- Inventory Management Techniques for Grocery Shops in Bangladesh — ResearchGate
- Internet subscriber growth outpaces mobile connections — Financial Express
- Mobile phone ownership rises: BBS — UNB
- BBS survey: household internet and smartphone access — Dhaka Tribune
- Bangla QR merchant figures — The Observer
- One QR, every wallet — The Daily Star

Social indicators

- 42% below fifth-grade education — Bonik Barta
- Functional literacy rate 62.92%: BBS — Prothom Alo
- Gender disparity rising in SME finance — Dhaka Tribune
- Digital lending for small businesses and merchants — TBS

Competitors

- Million shopkeepers using TallyKhata — TallyKhata
- Hishabee · Hishabee Impact Report 2025 · Hishabee Inventory Finance
- Startup Bangladesh invests in Hishab
- ShopUp debt financing — TBS

Costs, law & Grameenphone

- Gemini API pricing — Google
- Understand and count tokens — Google
- Supabase pricing 2026 — Toolradar
- Personal Data Protection Ordinance 2025: key takeaways — The Daily Star
- Key highlights of the PDPO 2025 for businesses — Mahbub & Company
- Grameenphone launches "AI & I" programme — The Daily Star
- Telenor Capital Markets Day 2017 — Grameenphone
- Grameenphone digital solution collaborates with PriyoShop — TBS